$$A = \begin{bmatrix} 2 & 2 & 1 \\ 0 & 1 & 2 \\ 3 & 4 & 0 \end{bmatrix}$$

$$\begin{array}{r} \frac{1}{2}\cdot \widetilde{F1} \\ \\ \\ \end{array} \left[\begin{array}{ccc|ccc} 1 & 1 & \frac{1}{2} & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 3 & 4 & 5 & 0 & 0 & 1 \end{array}\right]$$

$$\begin{array}{r} \\ \\ -3\cdot F1 + \widetilde{F3} \end{array} \left[\begin{array}{ccc|ccc} 1 & 1 & \frac{1}{2} & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & -1 & \frac{7}{2} & -3 & 0 & 1 \end{array}\right]$$

$$\begin{array}{r} -1\cdot F2 + \widetilde{F1} \\ \\ 1\cdot F2 + \widetilde{F3} \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{-3}{2} & 1 & -1 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & \frac{11}{2} & -3 & 1 & 1 \end{array}\right]$$

$$\begin{array}{r} \\ \\ \frac{2}{11}\cdot \widetilde{F3} \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{-3}{2} & 1 & -1 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{-3}{11} & \frac{2}{11} & \frac{2}{11} \end{array}\right]$$

$$\begin{array}{r} \frac{3}{2}\cdot F3 + \widetilde{F1} \\ -2\cdot F3 + \widetilde{F2} \\ \\ \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{13}{22} & \frac{-8}{11} & \frac{3}{11} \\ 0 & 1 & 0 & \frac{6}{11} & \frac{7}{11} & \frac{-4}{11} \\ 0 & 0 & 1 & \frac{-3}{11} & \frac{2}{11} & \frac{2}{11} \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} \frac{13}{22} & \frac{-8}{11} & \frac{3}{11} \\ \frac{6}{11} & \frac{7}{11} & \frac{-4}{11} \\ \frac{-3}{11} & \frac{2}{11} & \frac{2}{11} \end{bmatrix}$$

Sistemas de ecuaciones lineales

$$x + 2y - z = 3$$
$$0x - 5y + 3z = -4$$
$$3x + y + 2z = -1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -5 & 3 & -4 \\ 3 & 1 & 2 & -1 \end{array}\right] \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$$

$$\begin{array}{r} \\ \\ -3 \cdot F1 + \widetilde{F3} \end{array} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -5 & 3 & -4 \\ 0 & -5 & 5 & -10 \end{array}\right]$$

$$\begin{array}{r} \\ \frac{-1}{5} \cdot \widetilde{F2} \\ \\ \end{array} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & \frac{-3}{5} & \frac{4}{5} \\ 0 & -5 & 5 & -10 \end{array}\right]$$

$$\begin{array}{r} -2 \cdot F2 + \widetilde{F1} \\ \\ 5 \cdot F2 + \widetilde{F3} \end{array} \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{5} & \frac{7}{5} \\ 0 & 1 & \frac{-3}{5} & \frac{4}{5} \\ 0 & 0 & 2 & -6 \end{array}\right]$$

$$\begin{array}{r} \\ \\ \frac{1}{2} \cdot \widetilde{F3} \end{array} \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{5} & \frac{7}{5} \\ 0 & 1 & \frac{-3}{5} & \frac{4}{5} \\ 0 & 0 & 1 & -3 \end{array}\right]$$

$$\begin{array}{r} \frac{-1}{5} \cdot F3 + \widetilde{F1} \\ \frac{3}{5} \cdot F3 + \widetilde{F2} \\ \\ \end{array} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -3 \end{array}\right]$$

$$\begin{array}{c}\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -3 \end{array}\right]\end{array}\begin{array}{c} x \\ y \\ z \end{array}$$

$$\boxed{x=2 \quad y=-1 \quad z=-3}$$

Caso de solucion vacia o que no tiene solucion $0 \ 0 \ 0 \mid k$

$k \neq 0$

$$x + 0y - 5z = 15$$
$$0x + y + 8z = -26$$
$$3x + 2y + z = 5$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -5 & 15 \\ 0 & 1 & 8 & -26 \\ 3 & 2 & 1 & 5 \end{array}\right]$$

$$\begin{array}{r} \\ \\ -3 \cdot F1 + F3 \end{array}\left[\begin{array}{ccc|c} 1 & 0 & -5 & 15 \\ 0 & 1 & 8 & -26 \\ 0 & 2 & 16 & -40 \end{array}\right]$$

$$\begin{array}{r} \\ \\ -2 \cdot F2 + F3 \end{array}\left[\begin{array}{ccc|c} 1 & 0 & -5 & 15 \\ 0 & 1 & 8 & -26 \\ 0 & 0 & 0 & 12 \end{array}\right]$$

$$0 \ 0 \ 0 \mid k, \ k \neq 0$$

$$\boxed{S = \emptyset}$$

Caso de solucion infinita

$0 \quad 0 \quad 0 \mid 0$

$$x + 2y - 3z = 3$$
$$0x + y - 2z = 2$$
$$0x - 4y + 8z = -8$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 3 \\ 0 & 1 & -2 & 2 \\ 0 & -4 & 8 & -8 \end{array}\right]$$

$-2 \cdot F2 + \widetilde{F1}$

$4 \cdot F2 + \widetilde{F3}$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{array}{cc} \begin{array}{ccc} x & y & z \end{array} & \\ \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right] & \begin{array}{c} x \\ y \\ z \end{array} \end{array}$$

$$x + z = -1 \rightarrow x = -z - 1$$
$$y - 2z = 2 \rightarrow y = 2z + 2$$

$$S = \{ (\underbrace{-z-1}_{x}, \underbrace{2z+2}_{y}, z) \}$$

$z = 0 \qquad -0 - 1, \; 2 \cdot 0 + 2, \; 0$

$$(-1, 2, 0)$$

02-52-2024

1. [3 pts] Calcule $\overline{(i+3)} \cdot \frac{5i}{2-i}$.

$$\overline{i+3} \cdot \frac{5i}{2-i} \cdot \frac{2+i}{2+i}$$

a^2-b^2

$(a-b)(a+b)$

$$\overline{i+3} \cdot \frac{5i(2+i)}{2^2-i^2}$$

$$\overline{i+3} \cdot \frac{10i+5i^2}{4+1}$$

$$-i+3 \cdot \frac{10i-5}{5}$$

$$-i+3 \quad \frac{\cancel{5}(2i-1)}{\cancel{5}}$$

$$(-i+3)(2i-1)$$

$$-2i^2+i+6i-3$$

$$2+i+6i-3$$

$$\boxed{7i-1}$$

2. Sean $x = 3\,\text{cis}(110°)$, $y = 2\,\text{cis}(85°)$, $z = xy$. $z = xy$

(a) [2 pts] Calcule z en forma rectangular.

$$z = 3\,\text{cis}(110) \cdot 2\,\text{cis}(85)$$

$$6\,\text{cis}(195)$$

$$6[\cos(195) + \text{sen}(195)i]$$

$$6\cos(195) + 6\,\text{sen}(195)i$$

$$\boxed{z \quad \overbrace{\frac{-3\sqrt{6}-3\sqrt{2}}{2}}^{a} \quad \overbrace{\frac{-3\sqrt{6}+3\sqrt{2}}{2}}^{b}\, i}$$

θ

(b) [1 pto] Encuentre el argumento principal de z.

$$\tan^{-1}\left(\frac{\frac{-3\sqrt{6}+3\sqrt{2}}{2}}{\frac{-3\sqrt{6}-3\sqrt{2}}{2}}\right) = \boxed{15}$$